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(54) **RADON SENSOR DEVICE USING POLYHEDRAL-SHAPED IONIZATION CHAMBER**

RADON-SENSORVORRICHTUNG MIT POLYEDRISCHER IONISATIONSKAMMER

DISPOSITIF DE CAPTEUR DE RADON UTILISANT UNE CHAMBRE D'IONISATION DE FORME POLYÉDRIQUE

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Description

Technical Field

[0001] The present disclosure relates to a radon sensor using an ionization chamber and, more particularly, to a radon sensor device using a polyhedral-shaped ionization chamber.

Background Art

[0002] In general, an ionization chamber that is used for a radon sensor, as shown in FIG. 1, has a cylindrical structure in which a conductive cylinder having a plurality of pores and a straight probe unit disposed in the central axis in the cylinder are combined.

[0003] Such a cylindrical structure of the related art has the advantage that an electrical field intensifies toward the probe unit from the inner surface of the ionization chamber and is uniformly distributed in the axial direction, so the measurement efficiency is high. However, the cylindrical structure is difficult to mount and is vulnerable to electromagnetic wave noise due to the pores.

[0004] Further, there is inconvenience in that it is required to attach an appropriate filter to prevent dust from permeating through the pores. Further, since high-voltage bias is applied to the ionization chamber having a cylindrical structure and is completely exposed to the outside, there is also a possibility of leakage of electricity and an electric shock.

[0005] Further, air excessively smoothly flows inside due to the pores and thoron (Rn-220) having a short half-life of 55.6 seconds also flows into the ionization chamber, so there is a problem that there is a possibility of simultaneous measurement of radon (Rn-222) and thoron (Rn-220). WO9113371A1, US4853536A1, JP2005156463A and KR1020170023601A are examples of radon sensors.

Disclosure

Technical Problem

[0006] The present disclosure has been made in an effort to solve the problems described above and an objective of the present disclosure is to provide a radon sensor device using a polyhedral-shaped ionization chamber that has excellent mountability and mass productivity, resists well electromagnetic noise, has not possibility of leakage of electricity, has high measurement efficiency, and can exclude interference by thoron (Rn-220) instead of the cylindrical structure of the related art.

Technical Solution

[0007] In order to achieve the objectives, the present invention provides a radon sensor device using a polyhedral-shaped ionization chamber according to claim 1. The radon sensor device includes: an ionization chamber having an open side and inner sides surrounded by a first conductor and generating an electrical field therein by applying bias power to the first conductor; a cover having a first side covered with a second conductor and closing the open side of the ionization chamber such that that first conductor disposed on the inner sides of the ionization chamber and the second conductor are electrically connected; a probe unit disposed in the ionization chamber and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber; and a measurement circuit detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit into a predetermined magnitude.

[0008] The ionization chamber may be formed in a hexahedral shape using an insulating material.

[0009] The first conductor disposed on the inner sides of the ionization chamber may be formed by coating conductive paint with a predetermined thickness on the inner sides of the ionization chamber.

[0010] The bias power that is applied to the first conductor disposed on the inner sides of the ionization chamber is a DC voltage of 50V to 300V.

[0011] The first conductor disposed on the inner sides of the ionization chamber and the second conductor disposed on the first side of the cover may be electrically connected through joints between the cover and the ionization chamber.

[0012] The cover may be a printed circuit board (PCB) having a predetermined thickness and the entire of a first side of the printed circuit board may be covered with the second conductor.

[0013] A plurality of holes may be formed through the cover so that external air can flow into the ionization chamber or air can flow outside.

[0014] When a volume of the ionization chamber is 100cc, a diameter of the holes formed through the cover may be 0.2mm to 0.3mm and the number thereof may be 6 to 10.

[0015] The probe may detect ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber.

[0016] The probe unit is disposed at a center in the ionization chamber and is formed in a T-shape.

[0017] The probe unit is formed in a T-shape by perpendicularly combining first and second probes that have a rod shape or a straight shape having a predetermined length; and a first end of the first probe is electrically connected with the measurement circuit through the cover and a second end of the first probe is disposed at the center in the ionization chamber, and a center of the second probe is perpendicularly coupled to the second end of the first probe and is disposed at the center in the ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same.

[0018] The probe unit is disposed at a center in the

ionization chamber.

[0019] According to an example useful for understanding, but not forming part of, the presently claimed invention, the probe unit is formed by combining a first probe having a T-shape and second probes having a predetermined pattern, in which the first probe has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit through the cover; and the second probes are formed in a rod shape or a straight shape with a predetermined length, are coupled at centers to the upper head of the first probe, are longitudinally spaced a predetermined distance from each other, and are coupled perpendicular to or at a predetermined angle with respect to a longitudinal direction of the upper head of the first probe to have a predetermined pattern.

[0020] According to the present invention, the probe unit is formed by combining a first probe having a T-shape and second probes having a predetermined pattern, in which the first probe has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit through the cover; and the second probes are combined to vertically and horizontally cross each other at a right angle in a lattice pattern and centers thereof are longitudinally coupled to the upper head of the first probe.

[0021] The second probes may be coupled to the upper head of the first probe to vertically and horizontally cross each other at a right angle, whereby the lattice pattern thereof may be a square lattice pattern or a skew-crossed pattern.

[0022] The probe unit may be disposed in the ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same.

[0023] The measurement circuit may be disposed on a second side of the cover and may be mounted to be electrically connected with the second conductor disposed on the first side of the cover through a bias voltage outputter for outputting the bias power.

[0024] The measurement circuit may include: a guard ring installed such that the probe unit is disposed therethrough and absorbing and sensing a leakage current generated between the ionization chamber and the probe unit to a ground; a noise detector connected between the guard ring and the ground, sends a DC-type leakage current generated between the ionization chamber and the probe to the ground, and detects and outputs AC-type noise; first and second pre-amplifiers respectively connected to another end of the probe and an output end of the noise detector, and amplifying small electrical signals input from the probe and the noise detector into a predetermined magnitude; and a differential amplifier having a non-inverting terminal (+) and an inverting terminal (-) respectively connected to output ends of the first and second pre-amplifiers, and offsetting noise signals and outputting an alpha particle detection signal by amplifying a voltage difference of electrical signals pre-amplified through the first and second pre-amplifiers.

[0025] The noise detector may be configured by electrically connecting at least any one element of a resistance, a condenser, and a diode in series, in parallel, or in combination of series and parallel.

[0026] The radon sensor device may further include an electromagnetic wave-protective cap installed on a second side of the cover to surround the entire measurement circuit and is made of a conductive material to prevent signal disturbance of the measurement circuit due to external electromagnetic wave noise.

[0027] The electromagnetic wave-protective cap may be installed to be electrically connected with a ground of the measurement circuit.

Advantageous Effects

[0028] As described above, according to the radon sensor device using a polyhedral-shaped ionization chamber, it is possible to provide a radon sensor that has excellent mountability, resists well electromagnetic wave noise, is simply assembled with a low cost, and is suitable for mass production by using a polyhedral-shaped ionization chamber that has excellent mountability and mass productivity, resists well electromagnetic wave noise, has high measurement efficiency without a possibility of an electric shock or leakage of electricity, and can exclude interference of thoron (Rn-220) instead of an ionization chamber having a cylindrical shape of the related art.

Description of Drawings

[0029]

FIG. 1 is a perspective view showing an ionization chamber having a cylindrical structure that is applied to the related art;

FIG. 2 is an assembled perspective view showing a radon sensor device using a polyhedral-shaped ionization chamber according to an example useful for understanding, but not forming part of, the presently claimed invention.

FIG. 3 is an assembled cross-sectional view showing the radon sensor device using a polyhedral-shaped ionization chamber according to an example useful for understanding, but not forming part of, the presently claimed invention.

FIGS. 4a to 4d are perspective view showing a probe unit applied to another embodiment of the present invention; and

FIG. 5 is a detailed block diagram showing a measurement circuit applied to an embodiment of the present invention.

Best Mode

[0030] The objectives, characteristics, and advantages will be described in detail below with reference to the accompanying drawings, so those skilled in the art may easily achieve the scope of the present invention. In describing the present invention, detailed descriptions of well-known technologies will be omitted so as not to obscure the description of the present invention with unnecessary details.

[0031] Terms including ordinal numbers such as 'first', 'second', etc., may be used to describe various components, but the components are not to be construed as being limited to the terms. The terms are used only to distinguish one component from another component. For example, the 'first' component may be named the 'second' component, and vice versa, without departing from the scope of the present invention. Terms used in the present specification are used only to describe specific exemplary embodiments rather than limiting the present invention. Singular forms are intended to include plural forms unless the context clearly indicates otherwise.

[0032] Terminologies used herein were selected as terminologies that are currently used as generally as possible in consideration of the functions herein, but may be changed, depending on the intention of those skilled in the art, precedents, or advent of a new technology. Further, there are terminologies selected by applicant(s) at the applicant(s)' opinion in specific cases, and in these cases, the meanings will be described in the corresponding parts. Accordingly, the terminologies used herein should be defined on the basis of the meanings of the terminologies and the entire specification, not simply the names of the terminologies.

[0033] Throughout the present specification, unless explicitly described otherwise, "comprising" any components will be understood to imply the inclusion of other components rather than the exclusion of any other components. Further, the terms "~ part", "module", and the like mean a unit for processing at least one function or operation and may be implemented by hardware or software or by a combination of hardware and software.

[0034] Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings. However, the embodiments of the present invention that will be described hereafter may be modified in various ways and the scope of the present invention is not limited to the embodiments that will be described in detail below. Embodiments of the present invention are provided to more completely describe the present invention to those skilled in the art.

[0035] FIG. 2 is an assembled perspective view showing a radon sensor device using a polyhedral-shaped ionization chamber according to an example useful for understanding, but not forming part of, the presently claimed invention, FIG. 3 is an assembled cross-sectional view showing the radon sensor device using a polyhedral-shaped ionization chamber according to an example useful for understanding, but not forming part of, the presently claimed invention, FIGS. 4a to 4d are perspective views showing a probe unit applied to another embodiment of the present invention, and FIG. 5 is a detailed block diagram showing a measurement circuit applied to an embodiment of the present invention.

[0036] Referring to FIGS. 2 to 5, a radon sensor device using a polyhedral-shaped ionization chamber, includes an ionization chamber 100, cover 200, a support 300, a measurement circuit 400, etc. The polyhedral-shaped ionization chamber according to an embodiment of the present invention may further include an electromagnetic wave-protective cap 500, etc.

[0037] Hereafter, the components of the radon sensor device using a polyhedral-shaped ionization chamber according to the invention are described in detail.

[0038] The ionization chamber 100 has a polyhedral shape having an open side and inner sides surrounded by a first conductor 110 made of a conductive material, and generates an electrical field therein by applying bias power 10 to the first conductor 110.

[0039] It is preferable that the ionization chamber 100 is formed entirely in a hexahedral shape (e.g., a regular hexahedron or a rectangular parallelepiped) using an insulating material such as plastic, but is not limited thereto and may be a solid figure surrounded with four or more flat polygons such as a tetrahedron, a pentahedron, a heptahedron, and an octahedron.

[0040] Further, it is preferable that the first conductor 110 disposed on the inner sides of the ionization chamber 100 may be formed by coating conductive paint to have a predetermined thickness on the inner sides of the ionization chamber 100.

[0041] Further, it is preferable that the bias power 10 that is applied to the first conductor 110 disposed on the inner sides of the ionization chamber 100 may be a DC voltage of about 50V to 300V.

[0042] An electrical field is generated between the probe unit 300 in the ionization chamber 100, the first conductor 110 disposed on the inner sides of the ionization chamber 100, and a second conductor 210 disposed on a side of the cover 200 by applying high-voltage bias power 10 to the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210 disposed on a side of the cover 200 to be described below, whereby it is possible to absorb an ion current generated by alpha (α) decay through the probe unit 300. The ionization chamber 100 has a simple configuration, can be implemented at a low cost, and can perform 3-dimensional measurement.

[0043] Further, a stable DC voltage of about 50V to

300V (more preferably, about 50V to 150V) is used for the high-voltage bias power 10 that is applied to the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210 disposed on a side of the cover 200, thereby obtaining a condition in which additional ion charges can be effectively produced when alpha decay occurs in the ionization chamber 100. The power supplied from the high-voltage bias power 10 (e.g., a voltage or a current) may be changed in various ways, depending on the measurement range, the sensitivity, etc.

[0044] The cover 200 closes the open side of the ionization chamber 100, has a side covered with the second conductor 210 made of a conductive material, and electrically connects the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210, and is coupled such that the side on which the second conductor 210 is disposed and the open side of the ionization chamber 100 face each other, thereby closing the open side of the ionization chamber 100.

[0045] In this configuration, it is preferable that the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210 disposed on a side of the cover 200 are electrically connected through the joints C between the cover 200 and the ionization chamber 100.

[0046] The cover 200 may be a common printed circuit board (PCB) having a predetermined thickness and one entire side of the printed circuit board (PCB) may be covered with the second conductor 210.

[0047] Further, a plurality of 220 may be formed through the cover 200 so that external air can flow into the ionization chamber 100 or air can flow outside.

[0048] The diameter and the number of the holes 220 formed through the cover 200 may depend on the volume of the ionization chamber 100.

[0049] For example, when the volume of the ionization chamber is 100cc, it is preferable that the diameter is about 0.2mm to 0.3mm and the number is about 6 or 10.

[0050] In this case, since a diffusion time for external air to flow into the ionization chamber 100 is about 5 minutes to 10 minutes, the half-life of thoron (Rn-220) which is about 55.6 seconds is repeated about 5 times or more, so the concentration decreases to $1/2^5$ or less. Accordingly, interference by thoron (Rn-220) is excluded.

[0051] Further, since the holes 220 are small and large dust cannot flow inside, there is no need for a separate dust filter. Further, the open space of the ionization chamber 100 is minimized and the possibility to cause a measurement error due to permeation of electromagnetic wave noise is remarkably decreased in comparison to the ionization chamber having a cylindrical structure with several pores shown in FIG. 1.

[0052] Meanwhile, when the cover 200 is a printed circuit board (PCB), it is preferable that the holes 220 are pinholes of small defects or small holes formed through a film such as a metallic deposited film or an insulating film.

[0053] The probe unit 300 is made of a conductive material to be able to absorb ion charges produced when alpha decay occurs due to a radon (Rn-222) nuclide, etc. in the ionization chamber, is disposed in the ionization chamber 100, and absorbs ion charges produced when alpha (α) decay occurs in the ionization chamber 100.

[0054] That is, the probe unit 300 may be provided to detect ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber 100.

[0055] It is preferable that the probe unit 300 is provided to detect alpha particles produced when alpha decay occurs due to radon (Rn-222) gas from the air flowing in the ionization chamber 100, but it is not limited thereto and may be provided to detect all radioactive gases discharged as alpha (α) particles.

[0056] Further, the probe unit 300 is formed in a T-shape, as shown in FIGS. 2 and 3 as examples useful for understanding, but not forming part of, the presently claimed invention, so it is preferable that the probe unit 300 is disposed at the center in the ionization chamber 100.

[0057] That is, the probe unit 300 is formed in a T-shape by perpendicularly combine a first probe 300a and a second probe 300b that is formed in a rod shape or a straight shape with a predetermined length. A first end of the first probe 300a is electrically connected to the measurement circuit 400 through the cover 200 and a second end is disposed at the center in the ionization chamber 100.

[0058] Further, the center of the second probe 300b is perpendicularly coupled to the second end of the first probe 300a and is disposed at the center in the ionization chamber 100 such that the distances between the inner sides of the ionization chamber 100 and the surrounding portion of the second probe 300b are the same.

[0059] Further, the probe unit 300', as shown in FIGS. 4a to 4d as embodiments of the invention, may be formed in the shape of a common outdoor TV antenna and is preferably disposed at the center in the ionization chamber 100.

[0060] That is, as shown in FIGS. 4a and 4b, the probe unit 300' is formed by combining a T-shaped first probe 300a' and second probes 300b' having a predetermined pattern, and the first probe 300a' may be formed in the same shape as the probe unit 300 described above. That is, the first probe 300a' has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber 100, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to the center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit 400 through the cover 200.

[0061] Further, the second probes 300b' are formed in a rod shape or a straight shape with a predetermined length, are coupled at the centers to the upper head of the first probe 300a', and are longitudinally spaced a pre-

determined distance from each other. Further, the second probes 300b' may be coupled perpendicular to (see FIG. 4a) or at a predetermined angle θ (preferably, about 1 to 90 degrees) (FIG. 4b) with respect to the longitudinal direction of the upper head of the first probe 300a' in the same plane as the upper head of the first probe 300a', thereby having a predetermined pattern.

[0062] Further, as shown in FIGS. 4c and 4d, the probe unit 300' is formed by combining a T-shaped first probe 300a' and second probes 300b' having a predetermined pattern. The upper head of the first probe 300a' is formed in a rod shape or a straight shape with a predetermined length and is disposed at the center in the ionization chamber 100. The lower leg of the first probe 300a' is formed in a rod shape or a straight shape with a predetermined length and has a first end perpendicularly coupled to the center of the upper head and a second end electrically connected to the measurement circuit 400 through the cover 200.

[0063] Further, the second probes 300b' are combined to vertically and horizontally cross each other at a right angle in a lattice pattern and the centers thereof may be longitudinally coupled to the upper head of the first probe 300a'.

[0064] Further, the second probes 300b', as shown in FIG. 4c, are coupled to the upper head of the first probe 300a' to vertically and horizontally cross each other at a right angle in the same plane as the upper head of the first probe 300a', whereby the lattice pattern thereof may be a square lattice pattern. Further, as shown in FIG. 4d, the lattice pattern may be a skew-crossed pattern. Further, though not shown in the figures, for example, the lattice pattern may be formed in various shapes such as a rectangular lattice pattern, a square lattice pattern, or a changed lattice pattern.

[0065] That is, it is advantageous that the probe unit 300 or 300' is a T-shape or an outdoor TV antenna shape rather than a straight probe that is generally used in the ionization chamber having a cylindrical shape shown in FIG. 1 described above. The reason is because when a straight shape is applied to a polyhedral ionization chamber 100, the distance between the inner sides and the probe 300 or 300' is large and is not uniform, so it is difficult to maintain an intense electrical field in the ionization chamber 100.

[0066] Accordingly, the probe unit 300 or 300' should be designed such that the distance between the inner sides of the ionization chamber 100 and the probe unit 300 or 300' is maintained as small as possible and an intense electrical field can be generated around the probe unit 300 or 300'. When the probe unit 300 or 300' having a T-shape or an outdoor TV antenna shape proposed in embodiments of the present invention are designed, the distance from the inner sides of the ionization chamber 100 is small and the intensity of the electrical field around the probe unit 300 or 300' is also increased, whereby absorption efficiency when ion charges are produced is increased and accordingly the measurement efficiency can be increased.

[0067] That is, the probe unit 300 or 300' is designed to be disposed in the ionization chamber 100 such that the distances between the inner sides of the ionization chamber 100 and the surrounding portion of the probes are the same, whereby the distances between the surrounding portion of the probe unit 300 or 300' and the inner sides of the ionization chamber 100 are uniform and small and the intensity of the electrical field around the probe unit 300 or 300' is also increased. Accordingly, it is possible to effectively increase the measurement efficiency by increasing the absorption efficiency when ion charges are produced.

[0068] The probe unit 300 or 300' configured in this way is electrically connected to the measurement circuit 400 through the cover 200.

[0069] Further, the measurement circuit 400 outputs an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit 300 or 300' into a predetermined magnitude.

[0070] It is preferable that the measurement circuit 400 is disposed on a second side of the cover 200 and is mounted to be electrically connected with the second conductor 210 disposed on a first side of the cover 200 through a bias voltage outputter 401 for outputting bias power 10.

[0071] Further, the measurement circuit 400, as shown in FIG. 5, in a broad meaning, may include a guard ring 410, a noise detector 420, a first pre-amplifier 430, a second pre-amplifier 440, and/or a differential amplifier 450.

[0072] The guard ring 410 is made of a conductive material in a cylindrical shape, has the probe unit 300 or 300' disposed therethrough, and absorbs and sends a leakage current generated between the ionization chamber 100 and the probe unit 300 or 300' to a ground.

[0073] If the guard ring 410 is not provided, a current signal obtained from the probe unit 300 or 300' and a leakage current signal are combined, which causes the problem that the signal-to-noise ratio (SNR) decreases.

[0074] The noise detector 420 is connected between the guard ring 410 and the ground and maintains the potential of the guard ring 410 a little bit higher than the ground, thereby sending a DC-type leakage current generated between the ionization chamber 100 and the probe unit 300 or 300' to the ground and detecting and outputting an AC-type noise to the second pre-amplifier 440 to be described below.

[0075] It is preferable to configure the noise detector 420, for example, by electrically connecting at least any one element of a resistance, a condenser, and a diode in series, in parallel, and/or in combination of series and parallel.

[0076] The first pre-amplifier 430 is connected to a second end of the probe unit 300 or 300' and amplifies an electrical micro-signal input from the probe unit 300 or 300' into a predetermined magnitude.

[0077] The second pre-amplifier 440 is connected to an output end of the noise detector 420 and amplifies an

electrical micro-signal input from the noise detector 420 into a predetermined magnitude.

[0078] Output ends of the first pre-amplifier 430 and the second pre-amplifier 440 are respectively connected to a non-inverting terminal (+) and an inverting terminal (-) of the differential amplifier 450. The differential amplifier 450 can effectively offset noise signals and can detect an alpha particle detection signal by amplifying the electrical signals pre-amplified through the first pre-amplifier 430 and the second pre-amplifier 440 in proportion to the voltage difference of the electrical signals.

[0079] That is, as differential amplification is performed by the differential amplifier 450, it is possible to effectively offset noises having the same phase and flowing inside through the probe unit 300 or 300' and the noise detector 420 and it is possible to accurately and quickly obtain a high-sensitivity and low-noise alpha particle detection signal.

[0080] Further, the electromagnetic wave-protective cap 500 is installed on the second side of the cover 200 to surround the entire measurement circuit 400 and is made of a conductive material to prevent signal disturbance of the measurement circuit 400 due to external electromagnetic wave noise.

[0081] It is preferable that the electromagnetic wave-protective cap 500 is installed to be electrically connected with the ground of the measurement circuit 400.

[0082] Meanwhile, though not shown in the figures, a control unit (e.g., an MCU) that calculates a concentration value of alpha particles (e.g., a radon concentration value) by counting alpha particle detection signals (e.g., pulse signals), which are output from the differential amplifier 450, for a predetermined measurement time (e.g., 10 minutes, 30 minutes, and 1 hour), etc. may be provided.

[0083] That is, the present invention can effectively provide a radon sensor device using a polyhedral ionization chamber that has excellent mountability and mass productivity and can exclude interference of thoron (Rn-220).

[0084] Although preferred embodiments of a radon sensor device using a polyhedral ionization chamber according to the present invention were described above, the present invention is not limited thereto and may be modified in various ways without departing from claims, the detailed description, and the accompanying drawings, and they are also included in the present disclosure. The scope of the invention is defined by the appended claims.

Industrial Applicability

[0085] The present disclosure can be widely used in radon sensor devices.

Claims

1. A radon sensor device using a polyhedral-shaped ionization chamber, the radon sensor device comprising:

   said polyhedral-shaped ionization chamber (100) having an open side and inner sides surrounded by a first conductor (110) and for generating an electrical field therein by applying bias power to the first conductor (110);
   a cover (200) having a first side covered with a second conductor (210) and closing the open side of the ionization chamber (100) such that that first conductor (110) disposed on the inner sides of the ionization chamber (100) and the second conductor (210) are electrically connected;
   a probe unit (300) disposed in the ionization chamber (100) and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber (100); and
   a measurement circuit (400) detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit (300) into a predetermined magnitude, wherein
   the probe unit (300) is disposed at a center in the ionization chamber (100)
   characterized in that
   the probe unit (300') is formed by combining a first probe (300a') having a T-shape and second probes (300b') having a predetermined pattern, wherein the first probe (300a') has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit (400) through the cover (200); and
   the second probes (300b') are combined to vertically and horizontally cross each other at a right angle in a lattice pattern and centers thereof are longitudinally coupled to the upper head of the first probe.

2. The radon sensor device of claim 1, wherein the ionization chamber (100) is formed in a hexahedral shape using an insulating material.

3. The radon sensor device of claim 1, wherein the first conductor (110) disposed on the inner sides of the ionization chamber (100) is formed by coating conductive paint with a predetermined thickness on the inner sides of the ionization chamber (100).

4. The radon sensor device of claim 1, wherein the bias power that is applied to the first conductor (110) disposed on the inner sides of the ionization chamber is a DC voltage of 50V to 300V.

5. The radon sensor device of claim 1, wherein the first conductor (110) disposed on the inner sides of the ionization chamber (100) and the second conductor (210) disposed on the first side of the cover (200) are electrically connected through joints (C) between the cover (200) and the ionization chamber (100).

6. The radon sensor device of claim 1, wherein the cover (200) is a printed circuit board (PCB) having a predetermined thickness and the entire of a first side of the printed circuit board is covered with the second conductor (210).

7. The radon sensor device of claim 1, wherein a plurality of holes (220) is formed through the cover (200) so that external air can flow into the ionization chamber (100) or air can flow outside.

8. The radon sensor device of claim 7, wherein when a volume of the ionization chamber (100) is 100cc, a diameter of the holes (220) formed through the cover (200) is 0.2mm to 0.3mm and the number thereof is 6 to 10.

9. The radon sensor device of claim 1, wherein the probe unit (300) detects ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber (100).

10. The radon sensor device of claim 1, wherein the second probes (300b') are coupled to the upper head of the first probe (300a') to vertically and horizontally cross each other at a right angle, whereby the lattice pattern thereof is a square lattice pattern or a skew-crossed pattern.

11. The radon sensor device of claim 1, wherein the probe unit (300') is disposed in the ionization chamber (100) such that distances between the inner sides of the ionization chamber (100) and a surrounding portion of the second probe (300b') are the same.

12. The radon sensor device of claim 1, wherein the measurement circuit (400) is disposed on a second side of the cover (200) and is mounted to be electrically connected with the second conductor (210) disposed on the first side of the cover (200) through a bias voltage outputter for outputting the bias power.

13. The radon sensor device of claim 1, wherein the measurement circuit (400) comprises:

a guard ring (410) installed such that the probe unit (300) is disposed therethrough and absorbing and sensing a leakage current generated between the ionization chamber (100) and the probe unit (300) to a ground;
a noise detector (420) connected between the guard ring(410) and the ground, sends a DC-type leakage current generated between the ionization chamber (100) and the probe (300) to the ground, and detects and outputs AC-type noise;
first and second pre-amplifiers (430, 440) respectively connected to another end of the probe (300) and an output end of the noise detector (420), and amplifying small electrical signals input from the probe (300) and the noise detector (420) into a predetermined magnitude; and
a differential amplifier (450) having a non-inverting terminal (+) and an inverting terminal (-) respectively connected to output ends of the first and second pre-amplifiers (430, 440), and offsetting noise signals and outputting an alpha particle detection signal by amplifying a voltage difference of electrical signals pre-amplified through the first and second pre-amplifiers (430, 440).

14. The radon sensor device of claim 13, wherein the noise detector (420) is configured by electrically connecting at least any one element of a resistance, a condenser, and a diode in series, in parallel, or in combination of series and parallel.

15. The radon sensor device of claim 1, further comprising;

an electromagnetic wave-protective cap (500) installed on a second side of the cover (200) to surround the entire measurement circuit (400) and is made of a conductive material to prevent signal disturbance of the measurement circuit (400) due to external electromagnetic wave noise,
wherein the electromagnetic wave-protective cap (500) is installed to be electrically connected with a ground of the measurement circuit (400).

Patentansprüche

1. Radonsensorvorrichtung unter Verwendung einer polyedrisch geformten Ionisationskammer, wobei die Radonsensorvorrichtung umfasst:

die polyedrisch geformte Ionisationskammer (100), die eine offene Seite und Innenseiten aufweist, die durch einen ersten Leiter (110) umgeben sind, und zum Erzeugen eines elektri-

schen Felds darin durch Anlegen einer Vorspannungsleistung an den ersten Leiter (110);
eine Abdeckung (200), die eine erste Seite aufweist, die mit einem zweiten Leiter (210) bedeckt ist, und die offene Seite der Ionisationskammer (100) derart schließt, dass der erste Leiter (110), der an den Innenseiten der Ionisationskammer (100) angeordnet ist, und der zweite Leiter (210) elektrisch verbunden sind;
eine Sondeneinheit (300), die in der Ionisationskammer (100) angeordnet ist und Ionenladungen absorbiert, die erzeugt werden, wenn Alpha (α)-Zerfall in der Ionisationskammer (100) auftritt; und
eine Messschaltung (400), die ein Alphapartikel-Detektionssignal durch Verstärken und Verarbeiten eines elektrischen Mikrosignals, das von der Sondeneinheit (300) eingegeben wurde, in eine vorgegebene Größenordnung detektiert,
wobei die Sondeneinheit (300) in einem Zentrum in der Ionisationskammer (100) angeordnet ist,
dadurch gekennzeichnet, dass
die Sondeneinheit (300') durch Kombinieren einer ersten Sonde (300a'), die eine T-Form aufweist, mit zweiten Sonden (300b'), die ein vorgegebenes Muster aufweisen, gebildet ist,
wobei die erste Sonde (300a') eine oberen Kopf aufweist, der in einer Stabform oder einer geraden Form mit einer vorgegebenen Länge gebildet und im Zentrum in der Ionisationskammer angeordnet ist, einen unteren Schenkel aufweist, der in einer Stabform oder einer geraden Form mit einer vorgegebenen Länge gebildet ist und ein erstes Ende, das senkrecht mit einem Zentrum des oberen Kopfes verbunden ist, um eine T-Form zu bilden, und ein zweites Ende aufweist, das mit der Messschaltung (400) durch die Abdeckung (200) elektrisch verbunden ist; und
die zweiten Sonden (300b') kombiniert sind, um einander vertikal und horizontal in einem rechten Winkel in einem Gittermuster zu kreuzen, und Zentren von diesen in Längsrichtung mit dem oberen Kopf der ersten Sonde gekoppelt sind.

2. Radonsensorvorrichtung nach Anspruch 1, wobei die Ionisationskammer (100) in einer hexaedrischen Form unter Verwendung eines Isoliermaterials gebildet ist.

3. Radonsensorvorrichtung nach Anspruch 1, wobei der erste Leiter (110), der an den Innenseiten der Ionisationskammer (100) angeordnet ist, durch Beschichten von leitfähiger Farbe mit einer vorgegebenen Dicke auf den Innenseiten der Ionisationskammer (100) gebildet ist.

4. Radonsensorvorrichtung nach Anspruch 1, wobei die Vorspannungsleistung, die an den ersten Leiter (110) angelegt wird, der an den Innenseiten der Ionisationskammer angeordnet ist, eine DC-Spannung von 50 V bis 300 V ist.

5. Radonsensorvorrichtung nach Anspruch 1, wobei der erste Leiter (110), der an den Innenseiten der Ionisationskammer (100) angeordnet ist, und der zweite Leiter (210), der an der ersten Seite der Abdeckung (200) angeordnet ist, durch Verbindungsstellen (C) zwischen der Abdeckung (200) und der Ionisationskammer (100) elektrisch verbunden sind.

6. Radonsensorvorrichtung nach Anspruch 1, wobei die Abdeckung (200) eine Leiterplatte *(printed circuit board;* PCB) ist, die eine vorgegebene Dicke aufweist, und die Gesamtheit einer ersten Seite der Leiterplatte mit dem zweiten Leiter (210) bedeckt ist.

7. Radonsensorvorrichtung nach Anspruch 1, wobei eine Vielzahl von Löchern (220) durch die Abdeckung (200) gebildet ist, so dass Außenluft in die Ionisationskammer (100) strömen kann oder Luft nach außen strömen kann.

8. Radonsensorvorrichtung nach Anspruch 7, wobei, wenn ein Volumen der Ionisationskammer (100) 100 cc ist, ein Durchmesser der durch die Abdeckung (200) gebildeten Löcher (220) 0,2 mm bis 0,3 mm ist und die Anzahl von diesen 6 bis 10 ist.

9. Radonsensorvorrichtung nach Anspruch 1, wobei die Sondeneinheit (300) Ionenladungen detektiert, die durch eine Kollision von Luft und Alphapartikeln erzeugt wurden, die erzeugt werden, wenn Alpha (a)-Zerfall in der Ionisationskammer (100) auftritt.

10. Radonsensorvorrichtung nach Anspruch 1, wobei die zweiten Sonden (300b') mit dem oberen Kopf der ersten Sonde (300a') gekoppelt sind, um einander vertikal und horizontal in einem rechten Winkel zu kreuzen, wodurch das Gittermuster von diesen ein quadratisches Gittermuster oder ein schräg gekreuztes Muster ist.

11. Radonsensorvorrichtung nach Anspruch 1, wobei die Sondeneinheit (300') in der Ionisationskammer (100) derart angeordnet ist, dass Abstände zwischen den Innenseiten der Ionisationskammer (100) und einem umgebenden Abschnitt der zweite Sonde (300b') gleich sind.

12. Radonsensorvorrichtung nach Anspruch 1, wobei die Messschaltung (400) an einer zweiten Seite der Abdeckung (200) angeordnet ist und montiert ist, um mit dem zweiten Leiter (210), der an der ersten Seite der Abdeckung (200) angeordnet ist, durch eine Vor-

spannungsausgangseinrichtung zum Ausgeben der Vorspannungsleistung elektrisch verbunden zu sein.

13. Radonsensorvorrichtung nach Anspruch 1, wobei die Messschaltung (400) umfasst:

einen Schutzring (410), der derart installiert ist, dass die Sondeneinheit (300) durch diesen hindurch angeordnet ist, und einen Leckagestrom absorbiert und abtastet, der zwischen der Ionisationskammer (100) und der Sondeneinheit (300) zu einer Masse erzeugt wird;
einen Rauschdetektor (420), der zwischen dem Schutzring (410) und der Masse verbunden ist, einen DC-Leckagestrom sendet, der zwischen der Ionisationskammer (100) und der Sonde (300) zu der Masse erzeugt wird, und ein AC-Rauschen detektiert und ausgibt;
erste und zweite Vorverstärker (430, 440), die jeweils mit einem anderen Ende der Sonde (300) und einem Ausgangsende des Rauschdetektors (420) verbunden sind, und kleine elektrische Signale, die von der Sonde (300) und dem Rauschdetektor (420) eingegeben werden, in eine vorgegebene Größenordnung verstärken; und
einen Differenzverstärker (450), der einen nichtinvertierenden Anschluss (+) und einen invertierenden Anschluss (-) aufweist, die jeweils mit Ausgangsenden der ersten und zweiten Vorverstärker (430, 440) verbunden sind, und Rauschsignale offsettet und durch Verstärken einer Spannungsdifferenz von elektrischen Signalen, die durch die ersten und zweiten Vorverstärker (430, 440) vorverstärkt wurden, ein Alphapartikel-Detektionssignal ausgibt.

14. Radonsensorvorrichtung nach Anspruch 13, wobei der Rauschdetektor (420) durch elektrisches Verbinden wenigstens irgendeines Elements eines Widerstands, eines Kondensators und einer Diode seriell, parallel, oder in Kombination von seriell und parallel konfiguriert ist.

15. Radonsensorvorrichtung nach Anspruch 1, ferner umfassend:

eine Schutzkappe gegen elektromagnetische Wellen (500), die an einer zweiten Seite der Abdeckung (200) installiert ist, um die gesamte Messschaltung (400) zu umgeben, und aus einem leitenden Material besteht, um eine Signalstörung der Messschaltung (400) aufgrund von Rauschen externer elektromagnetischer Wellen zu verhindern,
wobei die Schutzkappe gegen elektromagnetische Wellen (500) installiert ist, um mit einer Masse der Messschaltung (400) elektrisch verbunden zu sein.

Revendications

1. Dispositif de capteur de radon utilisant une chambre d'ionisation de forme polyédrique, le dispositif de capteur de radon comprenant :

ladite chambre d'ionisation de forme polyédrique (100) ayant un côté ouvert et des côtés intérieurs entourés par un premier conducteur (110) et pour générer un champ électrique à l'intérieur en appliquant une puissance de polarisation au premier conducteur (110) ;
un couvercle (200) ayant un premier côté recouvert d'un deuxième conducteur (210) et fermant le côté ouvert de la chambre d'ionisation (100) de sorte que ce premier conducteur (110) disposé sur les côtés intérieurs de la chambre d'ionisation (100) et le deuxième conducteur (210) sont connectés électriquement ;
une unité de sondes (300) disposée dans la chambre d'ionisation (100) et absorbant des charges ioniques produites lorsqu'une désintégration alpha (α) se produit dans la chambre d'ionisation (100) ; et
un circuit de mesure (400) détectant un signal de détection de particules alpha en amplifiant et en traitant un micro-signal électrique entré par l'unité de sondes (300) en une amplitude prédéterminée, dans lequel
l'unité de sondes (300) est disposée à un centre de la chambre d'ionisation (100)
caractérisé en ce que
l'unité de sondes (300') est formée en combinant une première sonde (300a') ayant une forme en T et des deuxièmes sondes (300b') ayant un motif prédéterminé,
dans lequel la première sonde (300a') a une tête supérieure formée en forme de tige ou en forme rectiligne avec une longueur prédéterminée et est disposée au centre dans la chambre d'ionisation, a une branche inférieure formée en forme de tige ou en forme rectiligne avec une longueur prédéterminée et ayant une première extrémité connectée perpendiculairement à un centre de la tête supérieure pour former une forme en T et une deuxième extrémité connectée électriquement au circuit de mesure (400) à travers le couvercle (200) ; et
les deuxièmes sondes (300b') sont combinées pour se croiser verticalement et horizontalement les unes les autres à angle droit selon un motif en grillage et leurs centres sont couplés longitudinalement à la tête supérieure de la première sonde.

2. Dispositif de capteur de radon selon la revendication 1, dans lequel la chambre d'ionisation (100) est formée sous une forme hexaédrique à l'aide d'un matériau isolant.

3. Dispositif de capteur de radon selon la revendication 1, dans lequel le premier conducteur (110) disposé sur les côtés intérieurs de la chambre d'ionisation (100) est formé en revêtant de peinture conductrice d'une épaisseur prédéterminée les côtés intérieurs de la chambre d'ionisation (100).

4. Dispositif de capteur de radon selon la revendication 1, dans lequel la puissance de polarisation qui est appliquée au premier conducteur (110) disposé sur les côtés intérieurs de la chambre d'ionisation est une tension continue (CC) de 50 V à 300 V.

5. Dispositif de capteur de radon selon la revendication 1, dans lequel le premier conducteur (110) disposé sur les côtés intérieurs de la chambre d'ionisation (100) et le deuxième conducteur (210) disposé sur le premier côté du couvercle (200) sont connectés électriquement par l'intermédiaire de joints (C) entre le couvercle (200) et la chambre d'ionisation (100).

6. Dispositif de capteur de radon selon la revendication 1, dans lequel le couvercle (200) est une carte de circuit imprimé (PCB) ayant une épaisseur prédéterminée et la totalité d'un premier côté de la carte de circuit imprimé est recouverte du deuxième conducteur (210).

7. Dispositif de capteur de radon selon la revendication 1, dans lequel une pluralité de trous (220) est formée à travers le couvercle (200) de sorte que de l'air externe peut circuler dans la chambre d'ionisation (100) ou de l'air peut s'écouler à l'extérieur.

8. Dispositif de capteur de radon selon la revendication 7, dans lequel lorsqu'un volume de la chambre d'ionisation (100) est de 100 cm^3, un diamètre des trous (220) formés à travers le couvercle (200) est de 0,2 mm à 0,3 mm et leur nombre est de 6 à 10.

9. Dispositif de capteur de radon selon la revendication 1, dans lequel l'unité de sondes (300) détecte des charges ioniques produites par une collision d'air et de particules alpha produites lorsque la désintégration alpha (α) se produit dans la chambre d'ionisation (100).

10. Dispositif de capteur de radon selon la revendication 1, dans lequel les deuxièmes sondes (300b') sont couplées à la tête supérieure de la première sonde (300a') pour se croiser verticalement et horizontalement les unes les autres à angle droit, de sorte que le motif en grillage de celles-ci est un motif en grillage carré ou un motif croisé en biais.

11. Dispositif de capteur de radon selon la revendication 1, dans lequel l'unité de sondes (300') est disposée dans la chambre d'ionisation (100) de sorte que des distances entre les côtés intérieurs de la chambre d'ionisation (100) et une partie environnante de la deuxième sonde (300b') sont identiques.

12. Dispositif de capteur de radon selon la revendication 1, dans lequel le circuit de mesure (400) est disposé sur un deuxième côté du couvercle (200) et est monté pour être connecté électriquement au deuxième conducteur (210) disposé sur le premier côté du couvercle (200) par l'intermédiaire d'un dispositif de sortie de tension de polarisation destiné à délivrer en sortie la puissance de polarisation.

13. Dispositif de capteur de radon selon la revendication 1, dans lequel le circuit de mesure (400) comprend :

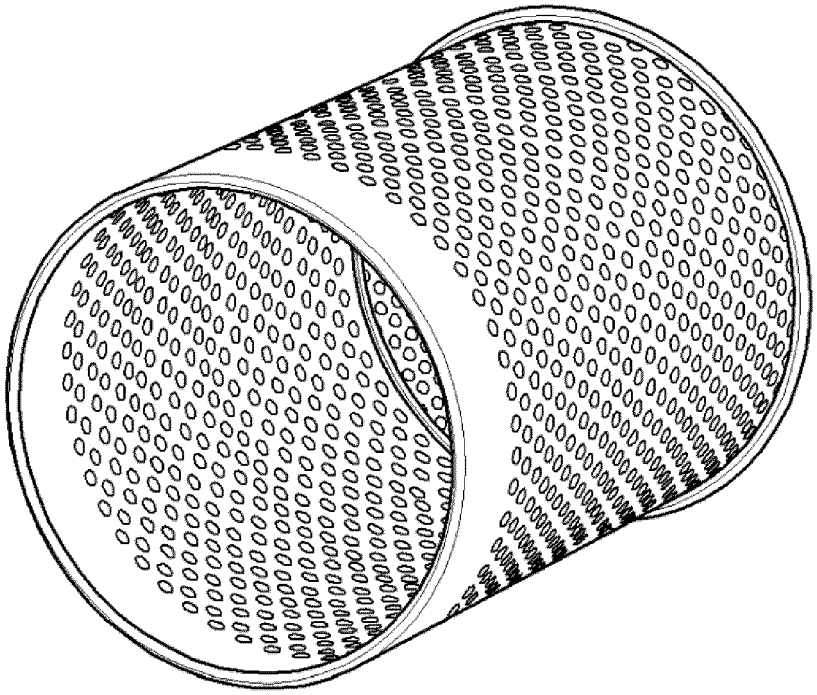
un anneau de protection (410) installé de telle sorte que l'unité de sondes (300) est disposée à travers celui-ci et absorbant et détectant un courant de fuite généré entre la chambre d'ionisation (100) et l'unité de sondes (300) vers une masse ;
un détecteur de bruit (420) connecté entre l'anneau de protection (410) et la masse, qui envoie un courant de fuite de type CC généré entre la chambre d'ionisation (100) et la sonde (300) vers la masse, et détecte et délivre en sortie un bruit de type CA ;
des premier et deuxième préamplificateurs (430, 440) connectés respectivement à une autre extrémité de la sonde (300) et à une extrémité de sortie du détecteur de bruit (420), et amplifiant des petits signaux électriques entrés par la sonde (300) et le détecteur de bruit (420) en une amplitude prédéterminée ; et
un amplificateur différentiel (450) ayant une borne non inverseuse (+) et une borne inverseuse (-) connectées respectivement à des extrémités de sortie des premier et deuxième préamplificateurs (430, 440), et décalant des signaux de bruit et délivrant en sortie un signal de détection de particules alpha en amplifiant une différence de tension de signaux électriques préamplifiés par les premier et deuxième préamplificateurs (430, 440).

14. Dispositif de capteur de radon selon la revendication 13, dans lequel le détecteur de bruit (420) est configuré en connectant électriquement au moins un élément quelconque d'une résistance, d'un condensateur, et d'une diode en série, en parallèle, ou en une combinaison série et parallèle.

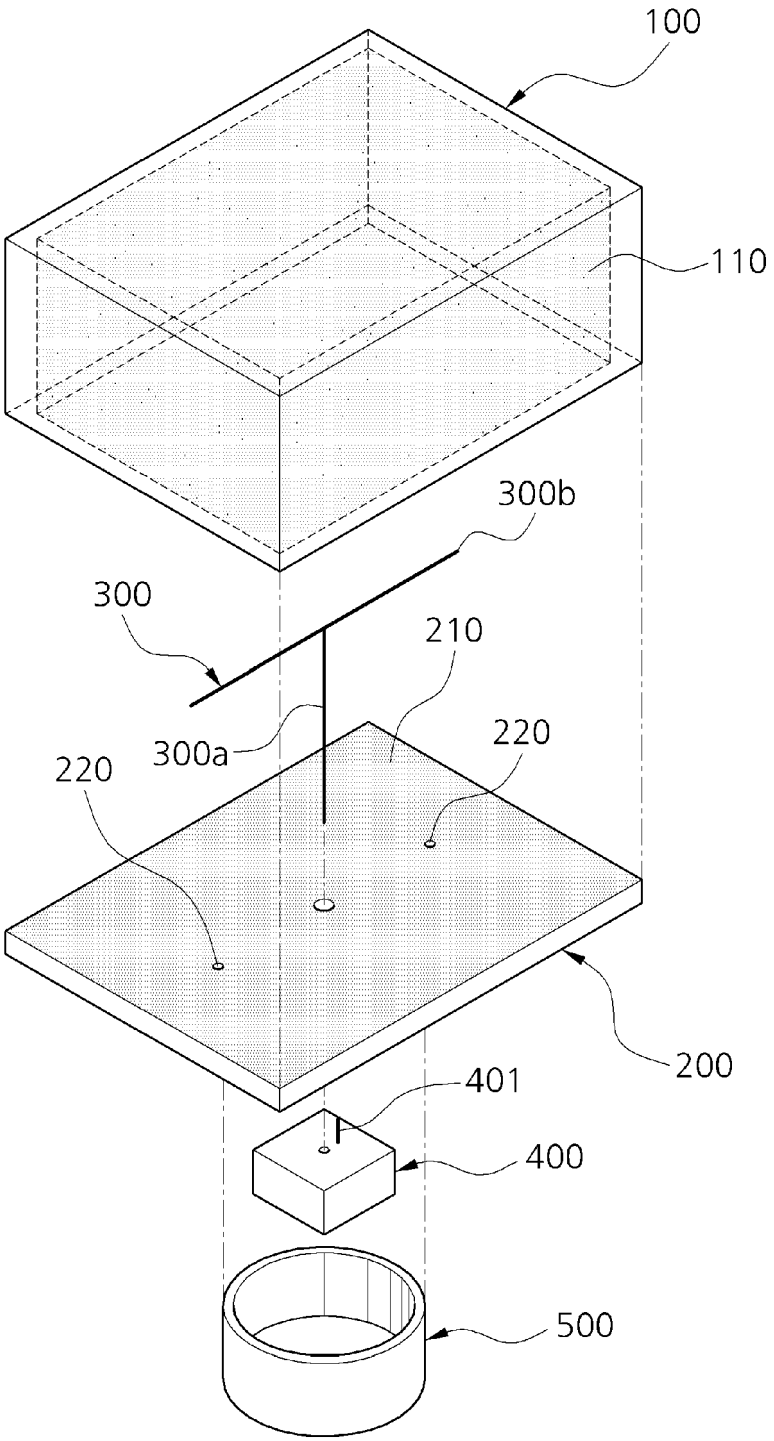
15. Dispositif de capteur de radon selon la revendication 1, comprenant en outre ;

un capuchon de protection contre les ondes électromagnétiques (500) installé sur un deuxième côté du couvercle (200) pour entourer l'ensemble du circuit de mesure (400) et constitué d'un matériau conducteur pour empêcher une perturbation de signal du circuit de mesure (400) due à un bruit par ondes électromagnétiques externes,
dans lequel le capuchon de protection contre les ondes électromagnétiques (500) est installé pour être connecté électriquement à une masse du circuit de mesure (400).

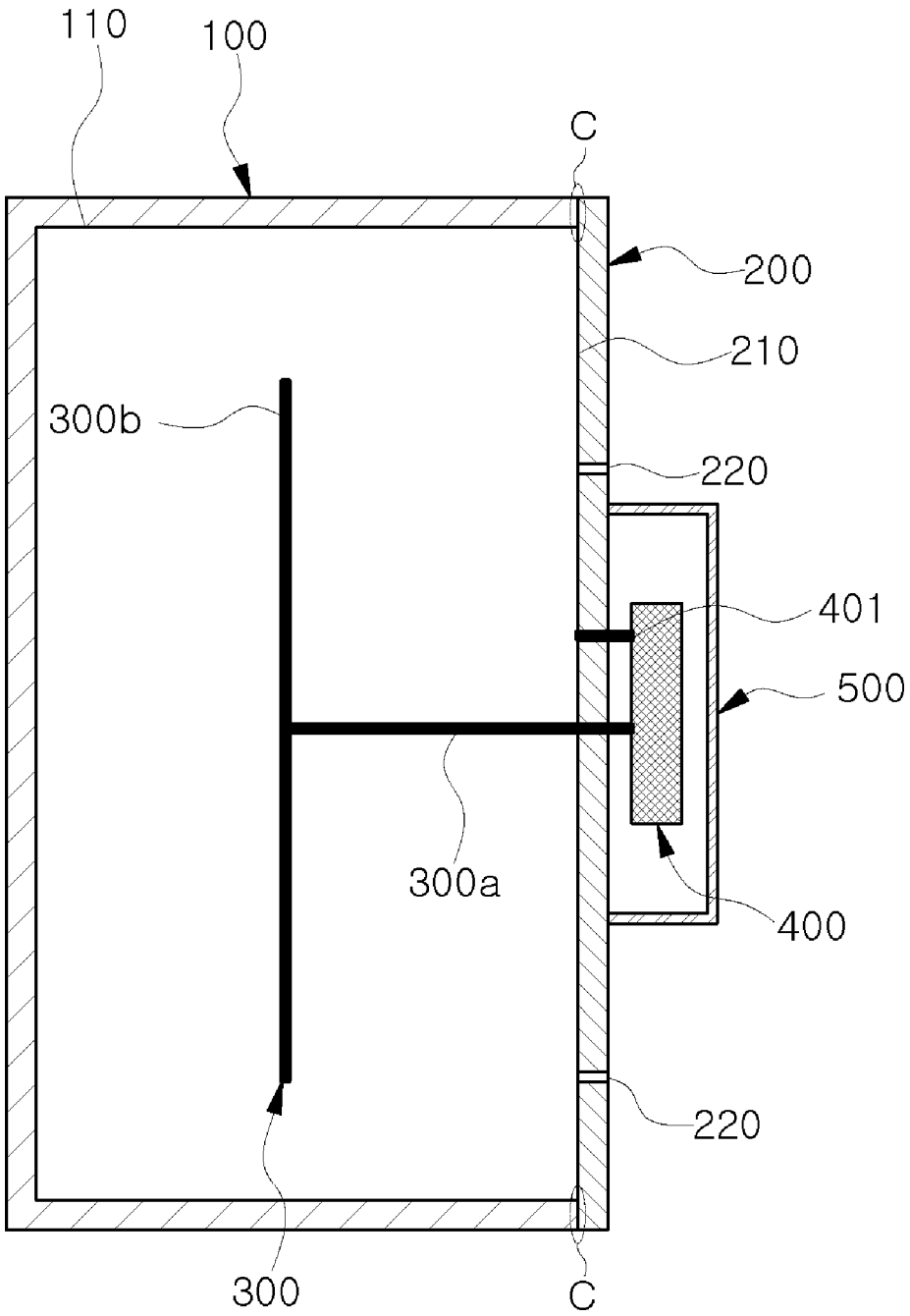
[FIG. 1]



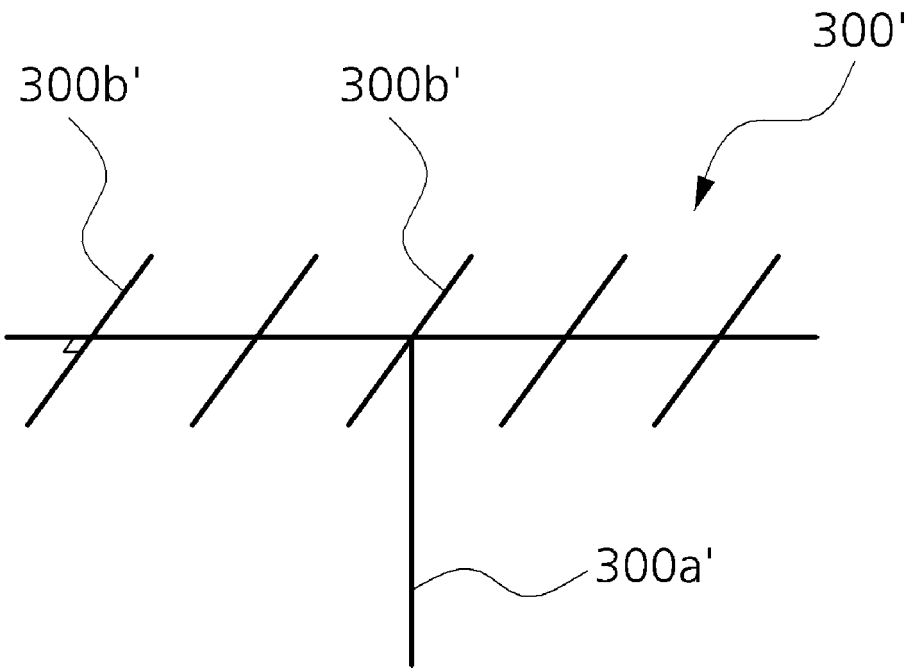
[FIG. 2]



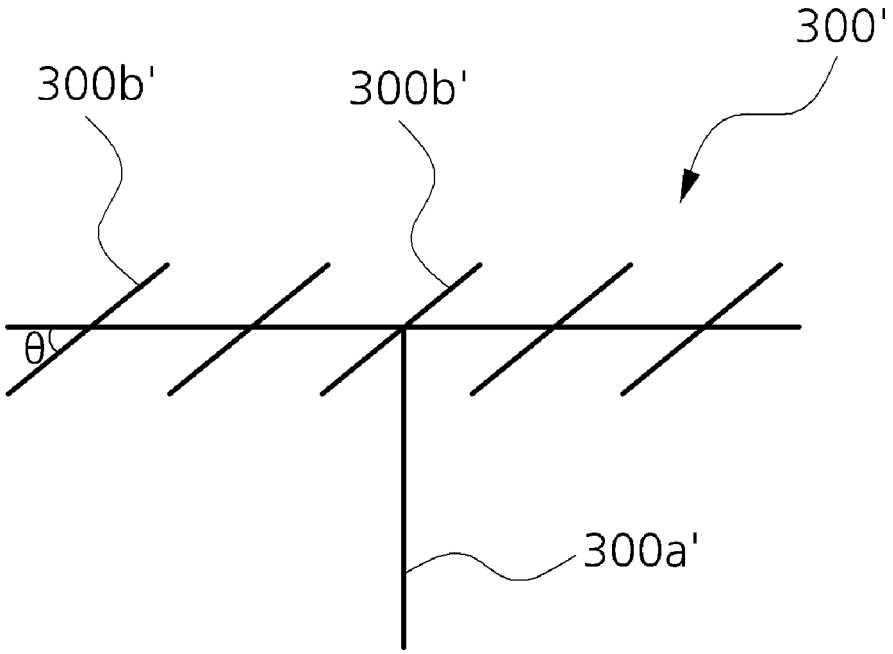
[FIG. 3]



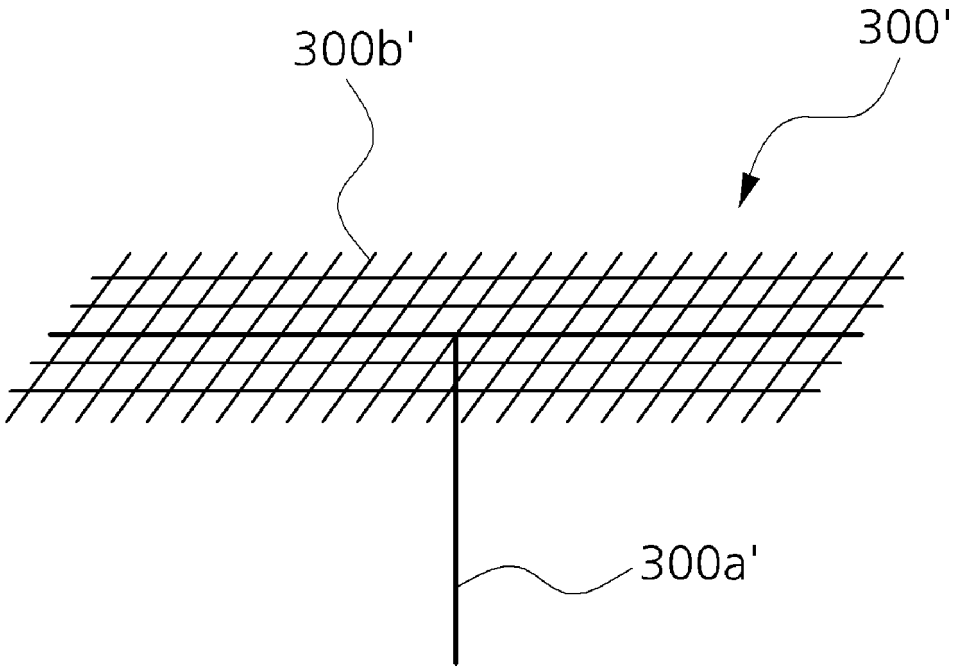
[FIG. 4A]



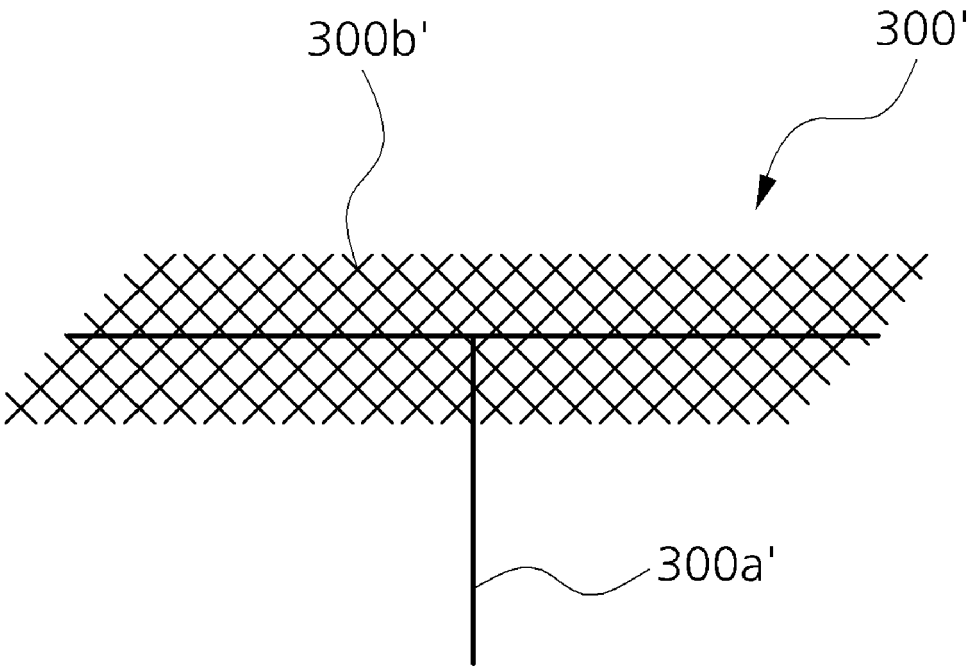
[FIG. 4B]



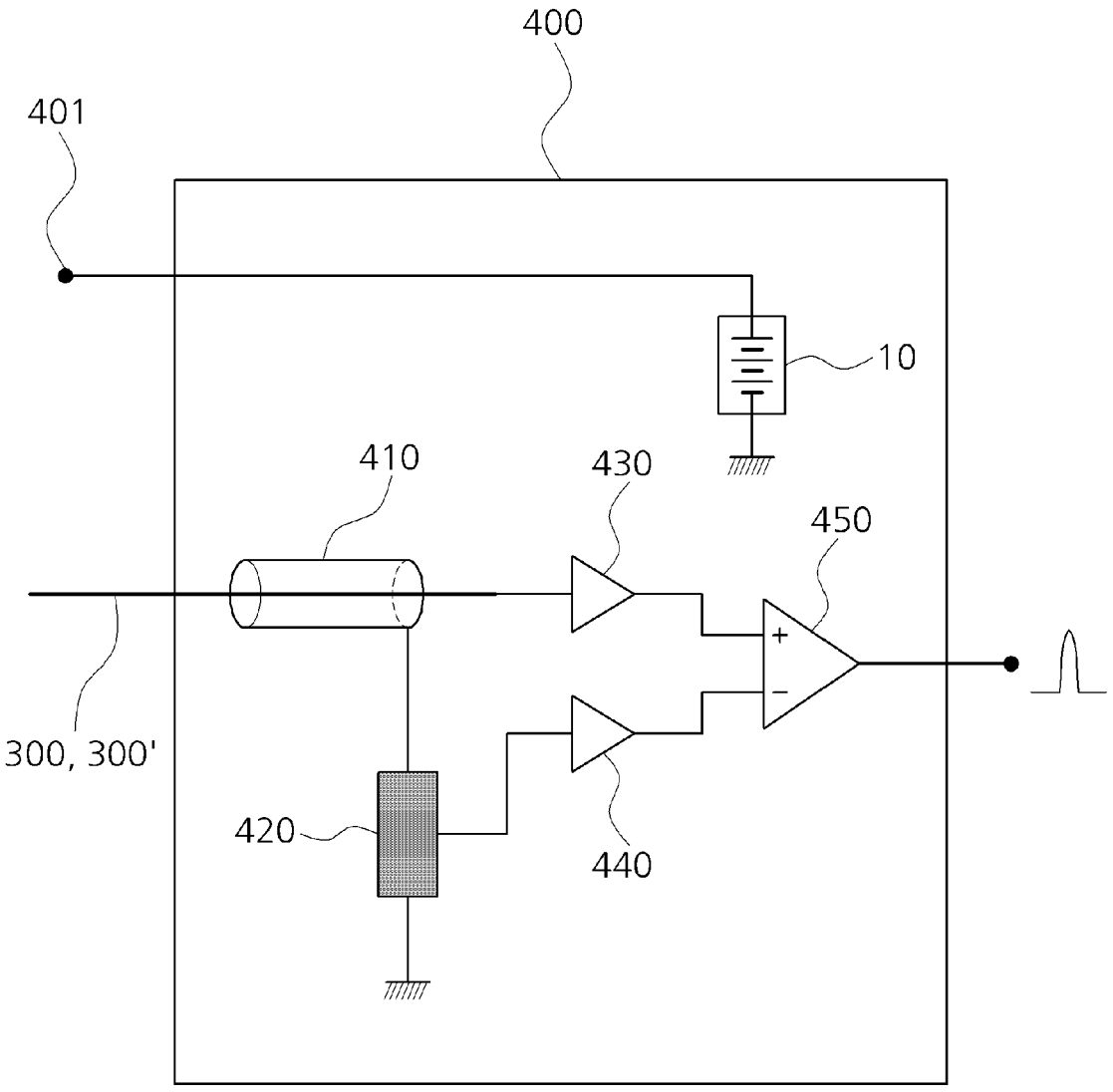
[FIG. 4C]



[FIG. 4D]



[FIG. 5]



REFERENCES CITED IN THE DESCRIPTION

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